

AIT304 ROBOTICS AND INTELLIGENT SYSTEMS

COMPREHENSIVE STUDY GUIDE

This study guide organizes topics according to the syllabus and highlights their importance based on question frequency in previous exams.

MODULE I: ROBOT CONFIGURATIONS AND CHARACTERISTICS

VERY IMPORTANT TOPICS (Repeatedly asked in exams)

1. Robot Configurations

- Four basic robot configurations according to coordinate systems (PPP, RPP, RRP, RRR)
- Cylindrical and spherical configurations
- Work envelope of robots

2. Robot Features and Dynamics

- General features of wheeled, legged, and aerial robots
- Dynamic characteristics of robots (speed of motion, load carrying capacity, speed of response)

3. Robot Laws and Applications

- Three laws of robotics
- Applications of robots

IMPORTANT TOPICS

1. Robot Anatomy

- Degrees of freedom
- Joint configurations
- Arm and body motion

2. Grippers and End Effectors

- Types of grippers (mechanical, magnetic, vacuum, adhesive)
- Active vs passive grippers
- Gripper force calculations

MODULE II: ACTUATORS AND SENSORS

VERY IMPORTANT TOPICS (Repeatedly asked in exams)

1. Sensor Characteristics

- Dynamic range, linearity, and resolution of sensors
- Difference between sensor and transducer

- Sensor selection criteria for robotics applications

2. Vision Sensors

- CCD vs. CMOS camera comparison
- Camera sensor hardware interfacing

3. Position and Motion Sensors

- Hall effect sensor working principle and applications
- Position and velocity internal sensors
- Touch sensors

4. Motors and Actuators

- Stepper motor vs. servomotor comparison
- DC motor control using H-Bridge and PWM
- Permanent magnet stepper motor operation

IMPORTANT TOPICS

1. Motion Control

- PID controller and its significance
- Mobile robot movement control
- Control drives

MODULE III: ROBOT VISION AND KINEMATICS

VERY IMPORTANT TOPICS (Repeatedly asked in exams)

1. Robot Vision Systems

- Seven stages of robot vision (sensing, pre-processing, segmentation, description, recognition, interpretation, feature extraction)
- Image processing analysis importance

2. Holonomic vs. Non-holonomic Robots

- Characteristics and differences
- Applications of each type

3. Omnidirectional Robots

- Configurations and setup
- Applications

IMPORTANT TOPICS

1. Robot Movement and Kinematics

- Differential drive significance

- Steerability and maneuverability
- Kinematic model of differential drive mobile robot

2. **Coordinate Frames and Transformations**

- Pure translation, pure rotation, and combined transformation
- Frame movement in space
- Mapping and transformation of coordinate systems
- Position and orientation representation

MODULE IV: LOCALIZATION AND MAPPING

VERY IMPORTANT TOPICS (Repeatedly asked in exams)

1. **Robot Localization**

- Mobile robot localization concept
- Localization problems and challenges
- Probabilistic map-based localization
- Kalman Filter method in localization

2. **SLAM (Simultaneous Localization and Mapping)**

- Mathematical definition and concept
- Applications and advantages
- Graph-based vs. particle SLAM
- Integration with wheel odometer data

IMPORTANT TOPICS

1. **Odometry and Error Modeling**

- Accurate odometry concept
- Error sources in odometric data
- Error model for odometric position estimation

2. **Map Representation**

- Decomposition strategies
- Current challenges in map representation

MODULE V: PATH PLANNING AND CONTROL

VERY IMPORTANT TOPICS (Repeatedly asked in exams)

1. **Path Planning Algorithms**

- Dijkstra's algorithm with examples (Most repeated question)

- Bug algorithm for obstacle avoidance
- A* and D* algorithms
- Potential field method

2. Control Decomposition

- Control decomposition concept
- Horizontal decomposition of robotic control systems
- Modularity for code reuse and sharing

IMPORTANT TOPICS

1. Neural Networks in Robotics

- Neural network approach in robot navigation
- Processing sensory data (cameras, Lidar)
- Training neural networks for navigation improvement

2. Obstacle Avoidance

- Bug algorithm assumptions
- Vector Field Histogram
- Dynamic window approaches (local vs. global)

3. Traversal Algorithms

- Breadth-first vs. depth-first search
- Comparison of traversal algorithms

TOP 10 MOST FREQUENTLY ASKED QUESTIONS

1. Dijkstra's algorithm with examples (Module V - Asked in 4 exams)

- 2023 June, 2024 Dec, 2024 Jan, 2024 May

2. Seven stages of robot vision (Module III - Asked in 3 exams)

- 2024 Dec, 2024 Jan, 2024 May

3. Hall effect sensor working principle (Module II - Asked in 3 exams)

- 2024 Dec, 2024 Jan, 2024 May

4. Holonomic vs. nonholonomic robots (Module III - Asked in 3 exams)

- 2024 Dec, 2024 Jan, 2024 May

5. Dynamic range, linearity, and resolution of sensors (Module II - Asked in 3 exams)

- 2024 Dec, 2024 Jan, 2024 May

6. Control decomposition concept (Module V - Asked in 3 exams)

- 2023 June, 2024 Dec, 2024 May

7. **Omnidirectional robot configurations** (Module III - Asked in 3 exams)
 - 2023 June, 2024 Dec, 2024 May
8. **Probabilistic map-based localization** (Module IV - Asked in 2 exams)
 - 2023 June, 2024 May
9. **Robot configurations according to coordinate systems** (Module I - Asked in 2 exams)
 - 2023 June, 2024 May
10. **CCD vs. CMOS Camera comparison** (Module II - Asked in 2 exams)
 - 2023 June, 2024 Dec

EXAM PREPARATION TIPS

1. **Focus on repeatedly asked questions** - Particularly Dijkstra's algorithm, seven stages of robot vision, and sensor characteristics.
2. **Practice numerical problems** - Especially in robot kinematics, gripper force calculations, and transformation matrices.
3. **Understand core concepts** - The distinction between holonomic and non-holonomic robots, SLAM, and path planning algorithms are fundamental.
4. **Draw diagrams** - For robot configurations, sensor working principles, and navigation algorithms.
5. **Review past question papers** - Pay special attention to Part B questions as they carry more marks.